1. Examples

We conclude with an example of the functionality – computing weight $(1,2)$ forms. Our ultimate goal is to produce non-CM examples of $(1,2)$ or $(1,1,2)$ forms and show that they correspond to geometry in some way (see [**?**] for a discussion of abelian fourfolds corresponding to forms of weight $(1,1,2)$). We are presently performing searches for such forms using the Hecke stability method discussed in **??**. As a byproduct, we have computed spaces of forms of weight $(2,3)$ and $(2,2,3)$ at hundreds of levels and nebentypii, and plan to eventually make these data available on the LMFDB. The codebase can be found here.

For now, we leave the reader with the following two examples.

1.1. A weight $(4,3)$ space over $\mathbb{Q}(\sqrt{2})$.

In this subsection, we reproduce a computation of [**?**].

Let $k := (4,3)$, $\mathfrak{N} = (\sqrt{2}+3)|7$, and χ the nontrivial ray class character unramified away from $\mathfrak{N}$ and the infinite places. Then, $S_k(\mathfrak{N},\chi)$ is two-dimensional. Choosing the generator $\delta := -2\sqrt{2}+4$, the coefficients of one of the two conjugate δ-normalized eigenforms in this space are given in Table 1. The coefficients of f lie in $K = \mathbb{Q}[x]/(x^4+24x^2+46)$.

We give a snippet of the code that we used to compute Table 1 – it runs in under 7 seconds on an 11th Gen Intel(R) Core(TM) i7-11800H @ 2.30 GHz. Of course, there is a lot going on under the hood! Nonetheless, we feel that one of the values of this work is that it makes such computations accessible even to users who do not wish to get their hands dirty.

```
// specify the field, level, weight, and nebentypus
F := QuadraticField(2);
ZF := Integers(F);
k := [4, 3];
N := Factorization(7*ZF)[1][1];
H := HeckeCharacterGroup(N, [1,2]);
chi := H.1;

// controls how many coefficients we compute
BOUND := 200;

// compute an basis of q-expansions spanning the corresponding space
// the basis will be over F'(chi) = F
M := GradedRingOfHMFs(F, BOUND);
Mk := HMFSpace(M, N, k, chi);
Sk := CuspFormBasis(Mk);

// the dimension of the cusp space is 2
assert #Sk eq 2;

// diagonalizing the Hecke action to produce an eigenbasis
eigs := Eigenbasis(Mk, Sk : P:=10);
```

1.2. Weight $(1,2)$ forms over $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{5})$.

This project was motivated by a proposal ([**?**]) for computing explicit examples of 4-folds of Mumford's type by computing Hilbert modular forms that are expected to be associated to them. These Hilbert modular forms are expected to be non-CM forms of weight $(1,1,2)$ and have coefficients satisfying certain rationality conditions. One also expects a similar geometric origin for non-CM forms of weight $(1,2)$ with coefficients satisfying similar rationality conditions. While do we not yet have any examples of non-CM forms, we report on some of our findings thusfar.

Given k, $\mathfrak{N}$, and χ, we can compute the space of CM forms $D_k(\mathfrak{N},\chi) \subseteq S_k(\mathfrak{N},\chi)$ by searching through the CM extensions of F with conductor dividing $\mathfrak{N}$ and looking for Hecke characters with an infinity type dependent on k and behavior at finite places determined by χ. Given such a Hecke character, one can produce explicit formulas for the Fourier coefficients of the corresponding automorphically induced Hilbert modular form. As such, we can verify whether a given form is CM or not by checking to see if it lands in the CM space.

TABLE 1. Some Fourier coefficients of an eigenform in $S_k(\mathfrak{N}, \chi)$ for k, $\mathfrak{N}$, and χ given above. Here, π is a totally positive generator of a prime ideal $\mathfrak{p}$, δ is the chosen generator for the different of $\mathbb{Z}_F$, and α is a generator for the coefficient field $K = \mathbb{Q}[x]/(x^4 + 24x^2 + 46)$ such that $\alpha^2 = 7\sqrt{2} - 12$.

$\mathrm{Nm}(\mathfrak{p})$	π	$\mathbf{a}_{\pi\delta^{-1}}(\mathbf{f})$
2	$-\sqrt{2}+2$	$-\alpha$
7	$-3\sqrt{2}+5$	$\frac{1}{7}(-2\alpha^3+6\alpha^2-31\alpha-5)$
7	$-\sqrt{2}+3$	$\frac{1}{7}(\alpha^3+26\alpha)$
9	3	$\frac{1}{7}(3\alpha^3+106\alpha)$
17	$-2\sqrt{2}+5$	$\frac{1}{7}(-8\alpha^2-222)$
17	$-4\sqrt{2}+7$	$\frac{1}{7}(3\alpha^3+120\alpha)$
23	$-\sqrt{2}+5$	$\frac{1}{7}(26\alpha^2+564)$
23	$-7\sqrt{2}+11$	$\frac{1}{7}(36\alpha^2+26)$
25	5	$\frac{1}{7}(2\alpha^3-74\alpha)$
31	$-5\sqrt{2}+9$	$\frac{1}{7}(13\alpha^3+30\alpha)$
31	$-3\sqrt{2}+7$	$\frac{1}{7}(-30\alpha^2-122)$
41	$-8\sqrt{2}+13$	$\frac{1}{7}(-90\alpha^2-562)$
41	$-2\sqrt{2}+7$	$\frac{1}{7}(-6\alpha^3-338\alpha)$
47	$-11\sqrt{2}+17$	$\frac{1}{7}(122\alpha^2+78)$
47	$-\sqrt{2}+7$	$-9\alpha^3-164\alpha$
71	$-7\sqrt{2}+13$	$\frac{1}{7}(-74\alpha^3-930\alpha)$
71	$-5\sqrt{2}+11$	$\frac{1}{7}(-2\alpha^3-530\alpha)$
73	$-2\sqrt{2}+9$	$24\alpha^2+302$
73	$-12\sqrt{2}+19$	$\frac{1}{7}(-27\alpha^3-198\alpha)$
79	$-15\sqrt{2}+23$	$\frac{1}{7}(-46\alpha^3-132\alpha)$
79	$-\sqrt{2}+9$	$\frac{1}{7}(47\alpha^3+942\alpha)$
89	$-4\sqrt{2}+11$	$\frac{1}{7}(-206\alpha^2-2262)$
89	$-10\sqrt{2}+17$	$\frac{1}{7}(65\alpha^3+1228\alpha)$
97	$-8\sqrt{2}+15$	$\frac{1}{7}(-234\alpha^2+734)$
97	$-6\sqrt{2}+13$	$\frac{1}{7}(51\alpha^3+1550\alpha)$
103	$-13\sqrt{2}+21$	$\frac{1}{7}(150\alpha^2+1296)$
103	$-3\sqrt{2}+11$	$\frac{1}{7}(114\alpha^3+2110\alpha)$
113	$-16\sqrt{2}+25$	$\frac{1}{7}(46\alpha^3+160\alpha)$
113	$-2\sqrt{2}+11$	$\frac{1}{7}(-10\alpha^3-316\alpha)$
121	11	$\frac{1}{7}(170\alpha^2+4602)$
127	$-7\sqrt{2}+15$	$\frac{1}{7}(96\alpha^2+130)$
127	$-9\sqrt{2}+17$	$\frac{1}{7}(-272\alpha^2-660)$
137	$-4\sqrt{2}+13$	$\frac{1}{7}(-10\alpha^2-50)$
137	$-14\sqrt{2}+23$	$\frac{1}{7}(-74\alpha^2-90)$
151	$-3\sqrt{2}+13$	$\frac{1}{7}(172\alpha^2-162)$
151	$-17\sqrt{2}+27$	$\frac{1}{7}(114\alpha^2-1404)$
167	$-23\sqrt{2}+35$	$\frac{1}{7}(172\alpha^3+902\alpha)$
167	$-\sqrt{2}+13$	$\frac{1}{7}(-398\alpha^2-4944)$
169	13	$\frac{1}{7}(-22\alpha^3-1006\alpha)$
191	$-13\sqrt{2}+23$	$\frac{1}{7}(11\alpha^3+216\alpha)$
191	$-7\sqrt{2}+17$	$10\alpha^3+76\alpha$
193	$-4\sqrt{2}+15$	$-43\alpha^3-632\alpha$
193	$-18\sqrt{2}+29$	$\frac{1}{7}(129\alpha^3+2682\alpha)$
199	$-9\sqrt{2}+19$	$\frac{1}{7}(62\alpha^2-1440)$
199	$-11\sqrt{2}+21$	$\frac{1}{7}(142\alpha^3+682\alpha)$

Theorem 1. *The weight* $(1,2)$ *forms over* $F = \mathbb{Q}(\sqrt{2})$ *and* $F = \mathbb{Q}(\sqrt{5})$ *of Galois stable level* $\mathfrak{N}$ *with* $\mathrm{Nm}(\mathfrak{N}) \leq 1500$ *and quadratic nebentypus character are all CM forms.*

It may seem odd to focus on Galois stable level. While there are indications that forms of Galois stable level might relate more easily to geometry, the reason in our setting is practical. For such levels $\mathfrak{N}$, one can check by a class field theory computation that unless $\mathfrak{N}|(2)$, there will always exist Eisenstein series $g \in M_{\underline{1}}(\mathfrak{N}, \psi)$ which are nonvanishing at the cusp at infinity. While the Hecke stability method will work with Eisenstein series of any weight, the bottleneck in the computation of $(1,2)$ forms is computing a basis of q-expansions for a space of weight $(1+k_g, 2+k_g)$, where k_g is the weight of the Eisenstein series.

References

[Cal21] Frank Calegari. Polymath proposal: 4-folds of Mumford's type, 2021.

[DLP19] Lassina Dembélé, David Loeffler, and Ariel Pacetti. Non-paritious Hilbert modular forms. *Math. Z.*, 292(1-2):361–385, 2019.